PROJECTION FROM A GEOMETRICALLY CONNECTED FACTOR INDUCES A BIJECTION ON CONNECTED COMPONENTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-varieties/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup, of the native theorem declarations with mathematical macros, and of the project's own no-op definitions of the reference display environments; no macro was invented or redefined.

Editorial note. All three referenced results live in earlier chapters, so their locators point at the verbatim lines of the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt) that carry the canonical tags.

Allowed standard prerequisites (editorial).

- Scheme theoretic fibres are homeomorphic to the set theoretic fibres. Schemes, Lemma labelled schemes-lemma-fibre-topological (canonical Stacks tag 01K1); invoked at proof lines 747-748. Supplies that the scheme theoretic fibres of the projection may be treated as the set theoretic ones; the connectedness statement about them is retained in the counted span.
- A morphism of schemes over a field is universally open. Morphisms, Lemma labelled morphisms-lemma-scheme-over-field-universally-open (canonical Stacks tag 0383); invoked at proof lines 749-751. Supplies openness of the projection, which is the topological hypothesis of the criterion applied at the end.
- Connected fibres over a base with connected components give a bijection on components. Topology, Lemma labelled topology-lemma-connected-fibres-connected-components (canonical Stacks tag 0378); invoked at proof lines 752-754. Supplies the criterion that finishes the proof once the fibres are connected and the map is open.

Source reference. Original lemma 01K1: sources/stacks-chow/tags.txt, source lines 2034–2034. Cross-chapter standard prerequisite invoked once at proof lines 747-748 to identify scheme theoretic with set theoretic fibres.

Source reference. Original lemma 0383: sources/stacks-chow/tags.txt, source lines 4222–4222. Cross-chapter standard prerequisite invoked once at proof lines 749-751 for openness of the projection; the fibre-connectedness argument is retained in the counted span.

Source reference. Original lemma 0378: sources/stacks-chow/tags.txt, source lines 4191–4191. Cross-chapter standard prerequisite invoked once at proof lines 752-754 as the topological criterion that concludes the proof; the verification of its hypotheses is retained in the counted span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let k be a field. Let X, Y be schemes over k. Assume X is geometrically connected over k. Then the projection morphism*

$$p : X \times_k Y \longrightarrow Y$$

induces a bijection between connected components.

Proof. The scheme theoretic fibres of p are connected, since they are base changes of the geometrically connected scheme X by field extensions. Moreover the scheme theoretic fibres are homeomorphic to the set theoretic fibres, see Schemes, Lemma 01K1. By Morphisms, Lemma 0383 the map p is open. Thus we may apply Topology, Lemma 0378 to conclude. □